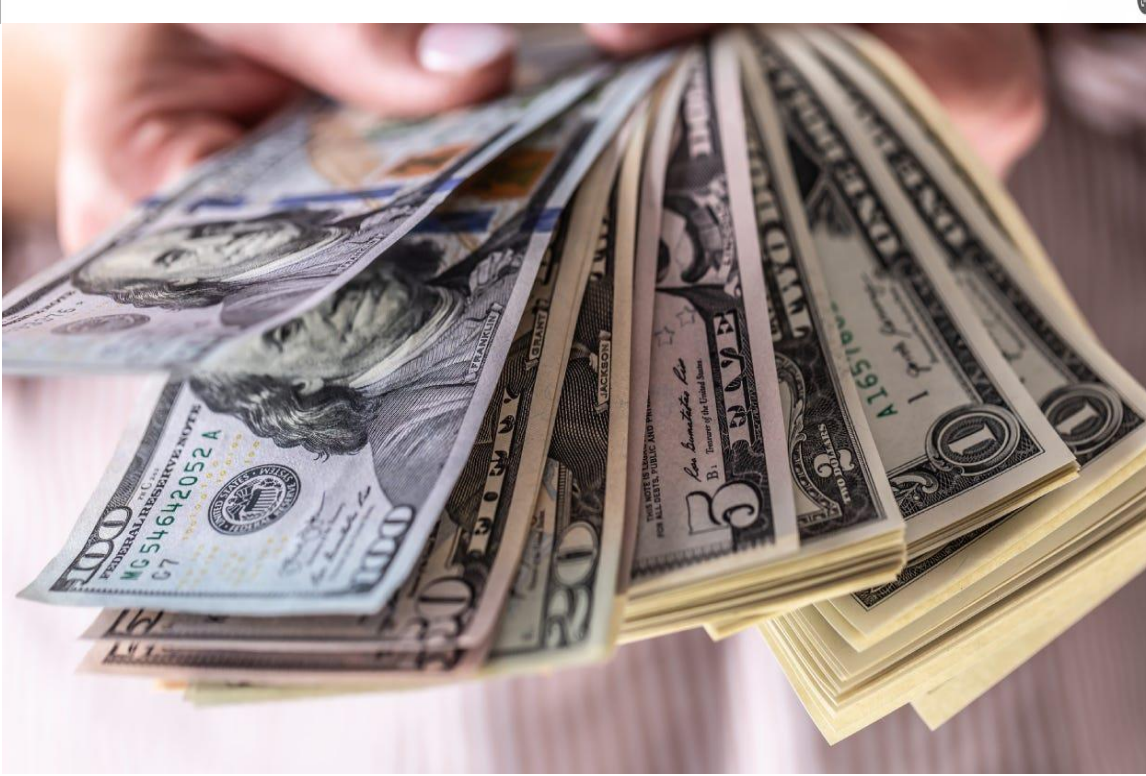
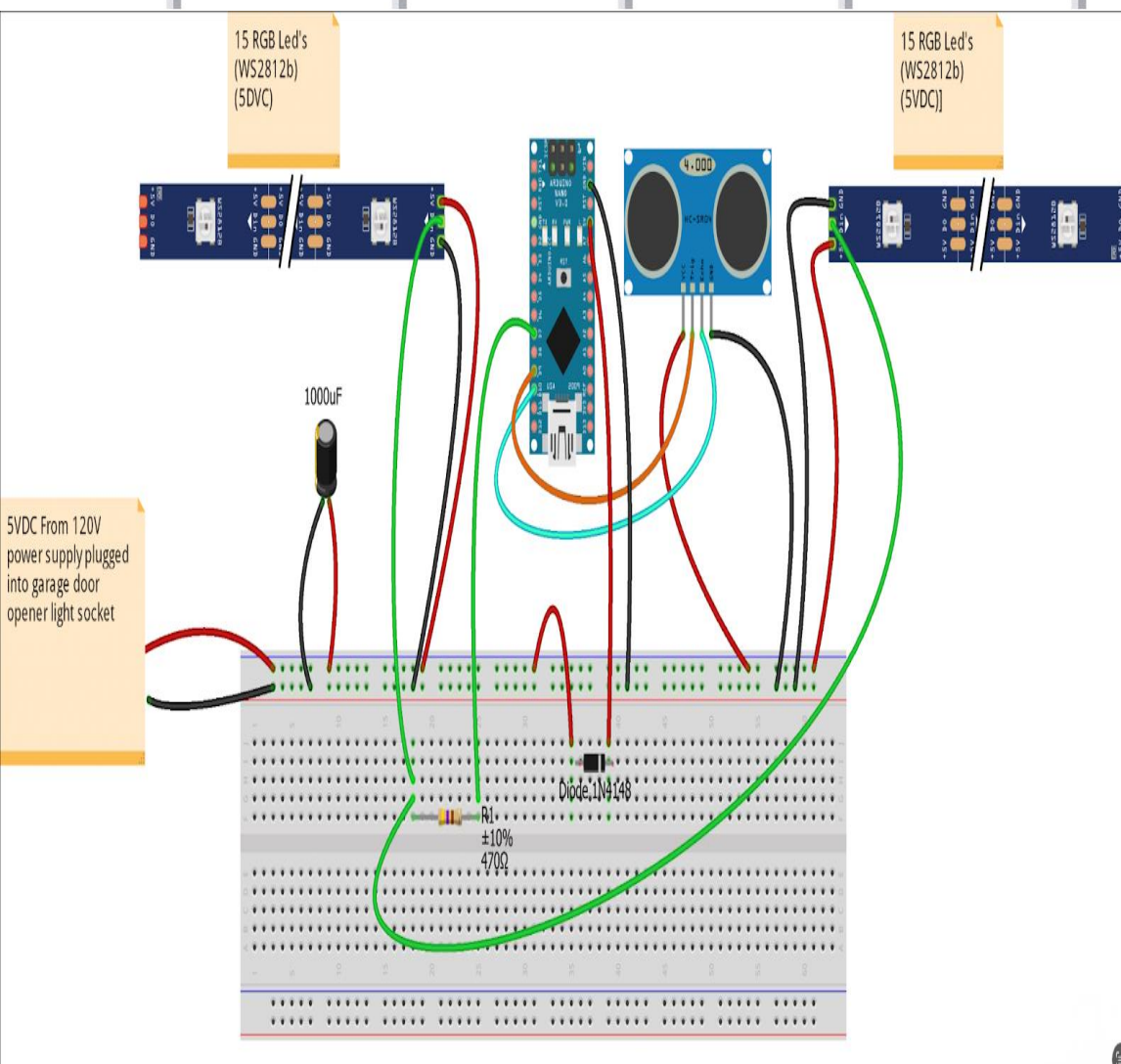
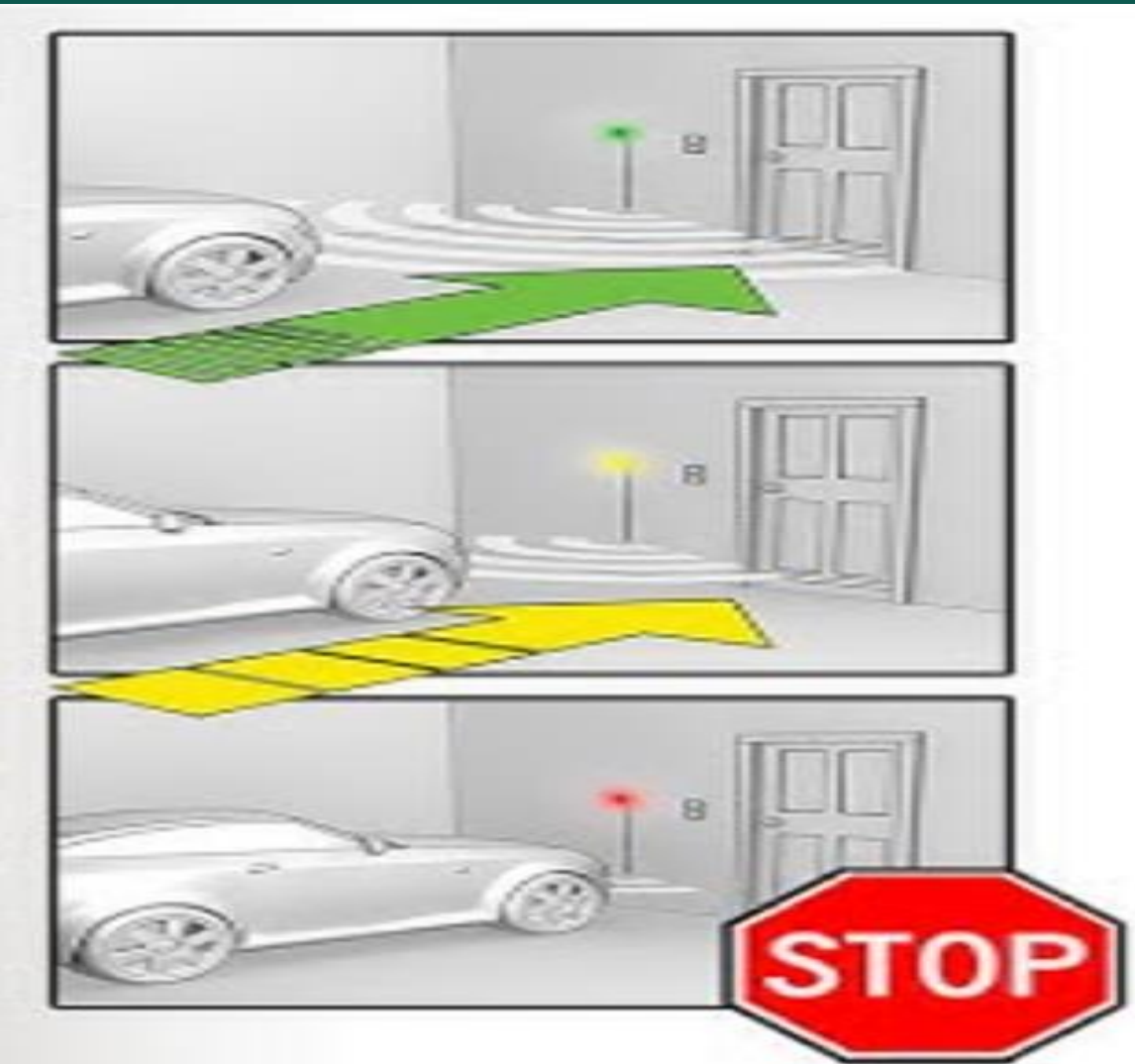


Garage Parking Sensor

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Professor Lubna Alazzawi



OPPORTUNITY/SIGNIFICANCE

Garage markers are unreliable. Our ultrasonic sensor with RGB LED feedback gives clear distance cues to improve safety, prevent damage, and save space.

TECHNICAL OBJECTIVES

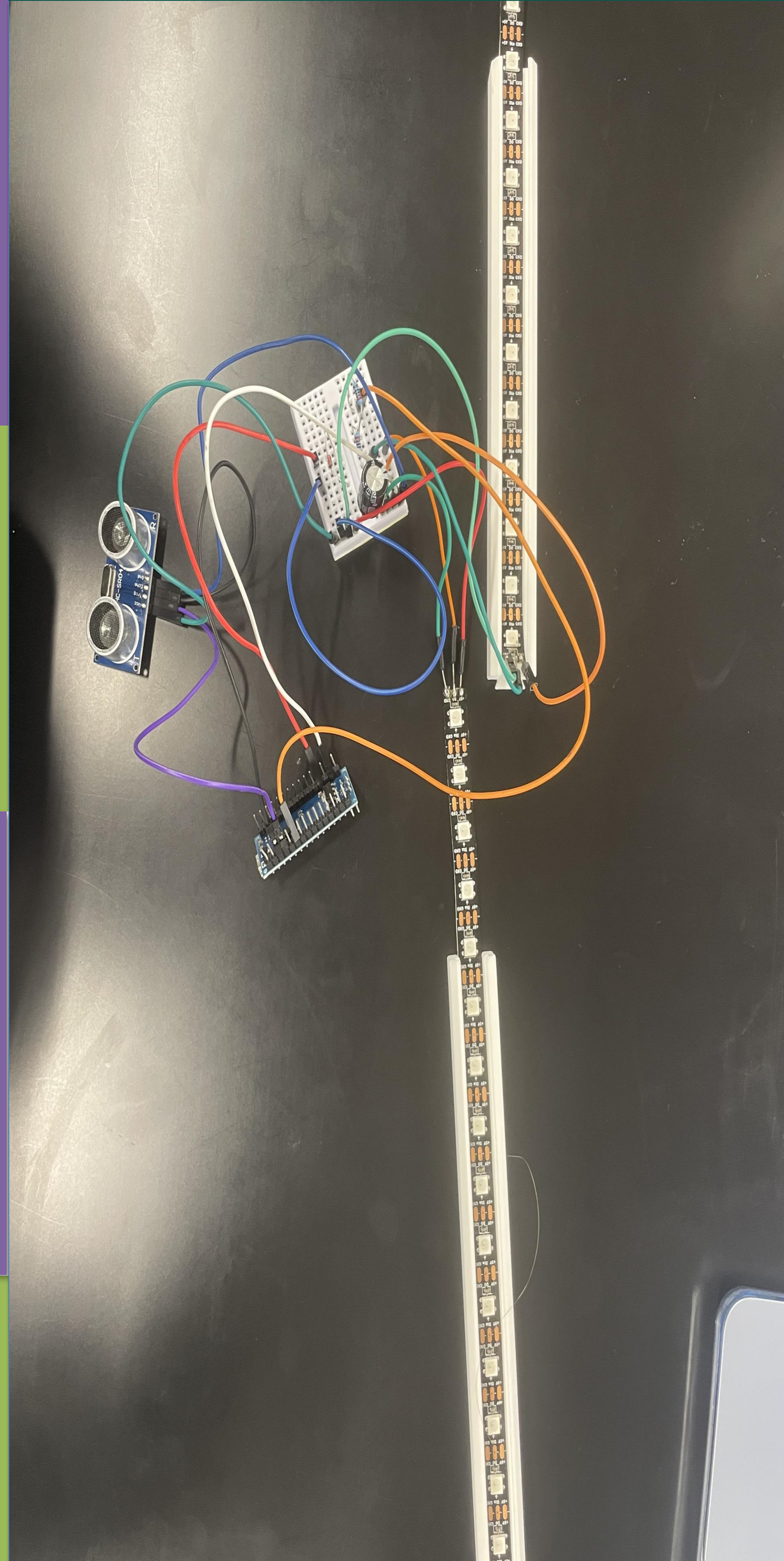
Create an accurate, affordable, easy-to-install distance sensor that gives fast, reliable LED feedback to guide drivers without a complicated setup.

ACCOMPLISHMENTS

We built a working prototype integrating the sensor, Arduino, and LED strip. Testing showed accurate readings and smooth color transitions. The full system cost about \$70 using readily available parts.

RELATED WORK

Other parking systems use costly, hard-to-install cameras or radar. Our improved ultrasonic-LED design is clearer, more reliable, and affordable for everyday garage use.



PROBLEM

Parking in garages is hard due to limited depth perception, causing dents and wasted space. Our Garage Parking Sensor gives clear visual distance cues to prevent these issues.

TECHNICAL APPROACH

We use an HC-SR04 ultrasonic sensor to measure distance and a WS2812B LED strip to display color-coded feedback. An Arduino processes the distance data and controls the LEDs. Supporting components ensure stable sensing and smooth color transitions. The system mounts on the wall or ceiling facing the vehicle.

NEXT STEPS

Test the device in real garage conditions (lighting, temperature, noise). Improve the housing for durability, refine wiring, and optimize the code for smoother LED transitions. Future upgrades may include wireless monitoring and cheaper production options.

COMMERCIALIZATION

- Phase 1:** Sell online to gather user feedback.
- Phase 2:** Partner with major home-improvement retailers to expand distribution.
- Phase 3:** Integrate with smart-home systems to increase long-term adoption.

